

Assignment Sheet

(Session: 2023-24)

Assignment No	string	Subject	string
Branch	string	Semester	string
Unit	Multiple Units	Faculty	string
Date Given	string	Submission Date	string

Q.No	Questions	CO	Marks
Q1	Explain the concept of stationarity in time series data and the role of differencing in achieving stationarity. Provide an example where differencing would be necessary	CO1	10
Q2	Derive the formula for the Mean Absolute Error (MAE) in the context of evaluating time series forecasting models and discuss its advantages and disadvantages compared to other metrics like Mean Squared Error (MSE)	CO1	10
Q3	Compare and contrast Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks in processing sequential data. Justify which network might be more suitable for handling long-term dependencies in time series forecasting and explain why	CO1	10
Q4	Evaluate the impact of seasonality on time series analysis. Using Holt-Winters Seasonal Model, demonstrate how to incorporate seasonality into forecasting and explain the intuition behind using multiplicative seasonal adjustments	CO1	10
Q5	Discuss the concept of sequence padding in the context of sequence models. Provide an example of a scenario where sequence padding is crucial and justify the choice of padding strategy used	CO1	10

Department of Artificial Intelligence

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Q6	Analyze the importance of feature engineering in prediction models. Provide an example of relevant features that can be extracted from a dataset for a regression problem and explain how they might influence the model's performance	CO1	10
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